

Predicting energy cost from wearable sensors: A dataset of energetic and physiological wearable sensor data from healthy individuals performing multiple physical activities

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Experimental Data Collection Methods

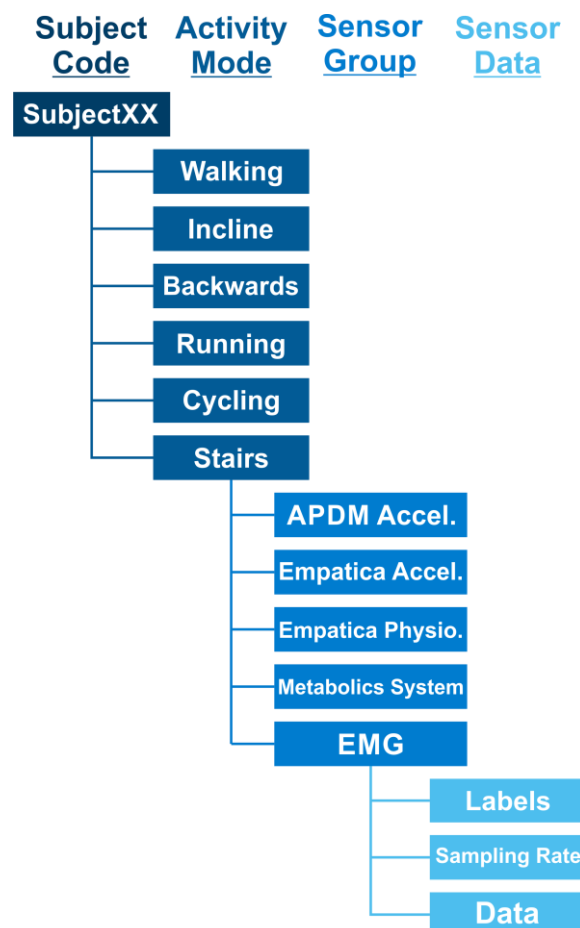
For a full description of the experimental methods please see the published article:

Ingraham, Kimberly A., Daniel Perry Ferris, and C. David Remy. "Evaluating Physiological Signal Salience for Estimating Metabolic Energy Cost from Wearable Sensors." *Journal of Applied Physiology* (2019). DOI: 10.1152/jappphysiol.00714.2018

File Structure

This dataset comprises ten (10) Matlab .mat files. Each .mat file contains *all* the raw sensor data for one individual subject; each is named according to the convention 'SubjectXX.mat'.

Each SubjectXX.mat file contains a data struct with the following fields:



Sensor Groups		Signals Contained in Data Matrix	
EMG (16 signals)	Left EMG (V) (8 signals)	Gluteus maximus Rectus femoris Vastus lateralis Semitendinosus Biceps femoris Medial gastrocnemius Soleus Tibialis anterior	
	Right EMG (V) (8 signals)	Gluteus maximus Rectus femoris Vastus lateralis Semitendinosus Biceps femoris Medial gastrocnemius Soleus Tibialis anterior	
APDM Accel (54 signals)	Waist (9 signals)	Acceleration (m/s^2)	x, y, z
		Angular Velocity (rad/s)	x, y, z
		Magnetic Field (uT)	x, y, z
	Chest (9 signals)	Acceleration (m/s^2)	x, y, z
		Angular Velocity (rad/s)	x, y, z
		Magnetic Field (uT)	x, y, z
	Left Ankle (9 signals)	Acceleration (m/s^2)	x, y, z
		Angular Velocity (rad/s)	x, y, z
		Magnetic Field (uT)	x, y, z
	Right Ankle (9 signals)	Acceleration (m/s^2)	x, y, z
		Angular Velocity (rad/s)	x, y, z
		Magnetic Field (uT)	x, y, z
	Left Foot (9 signals)	Acceleration (m/s^2)	x, y, z
		Angular Velocity (rad/s)	x, y, z
		Magnetic Field (uT)	x, y, z
	Right Foot (9 signals)	Acceleration (m/s^2)	x, y, z
		Angular Velocity (rad/s)	x, y, z
		Magnetic Field (uT)	x, y, z

Table 1: Signals Contained in each Sensor

Empatica Accel (6 signals)	Left Wrist (3 signals)	Acceleration (m/s ²)	x, y, z
	Right Wrist (3 signals)	Acceleration (m/s ²)	x, y, z
Empatica Physiological (4 signals)	Left Wrist (2 signals)	Electrodermal Activity (uS) Skin Temperature (degC)	
	Right Wrist (2 signals)	Electrodermal Activity (uS) Skin Temperature (degC)	
Metabolics System (7 signals)		VO2 (mL/s)	
		VCO2 (mL/s)	
		Respiratory Exchange Ratio (RER)	
		Breath Frequency (1/min)	
		Minute Ventilation (L/min)	
		Oxygen Saturation SpO2 (%)	
		Heart Rate (1/min)	

Table 1: Signals Contained in each Sensor Group (Continued)

Time vector (varies based on sampling rate)		Indicates activity performed (see Table 3)		Signal Data from a Signal Group (see Table 1)		
Contained in Labels variable	Time	Activity Code	Signal 1	Signal 2	...	Signal N
	0.001	22	0.0007	-0.0023	...	0.0007
	0.002	22	0	-0.003	...	0.0015
	0.003	22	-0.0053	-0.0015	...	0.0015
	0.004	22	-0.0083	-0.0023	...	0.0015
	0.005	22	-0.0075	-0.0015	...	0.0015
	0.006	22	-0.0053	-0.0023	...	0.0015
	0.007	22	-0.0046	-0.003	...	0.0015
	0.008	22	-0.0053	-0.0008	...	0.0007
	0.009	22	-0.0046	-0.0015	...	0.0022
	0.01	22	-0.0008	-0.0008	...	0.0015
	0.011	22	0.0015	-0.0008	...	0
	0.012	22	0.0067	-0.0008	...	0
	0.013	22	0.009	-0.0008	...	-0.0008
	0.014	22	0.009	-0.0008	...	0
	0.015	22	0.009	0	...	0.0007
Contained in Data variable	0.016	22	0.012	0.0015	...	0.0007
	0.017	22	0.0135	-0.0008	...	-0.0008
	0.018	22	0.0082	-0.0015	...	-0.0008
	.	.	.	.	...	.
	.	.	.	.	...	.
	.	.	.	.	...	.
	360.001	19	-0.0075	-0.003	...	0
	360.002	19	-0.0135	-0.0015	...	0.0015
	360.003	19	-0.0181	-0.003	...	0.0007
	360.004	19	-0.0256	-0.0015	...	0.0022
	360.005	19	-0.0293	-0.0008	...	0.003
	360.006	19	-0.024	-0.0015	...	0.0022
	360.007	19	-0.0165	-0.003	...	0.0015
	360.008	19	-0.0068	-0.003	...	0.0007
	360.009	19	0	-0.0015	...	-0.0015
	360.01	19	0.0045	-0.003	...	-0.0008

Note: If sensor data for a subject is missing (i.e., was not collected), the data reads NaN

Table 2: Example Data Matrix for a Sensor Group

Activity Code	Activity Mode	Intensity	Speed
1	Walking	-	0.6 (m/s)
2	Walking	-	0.9 (m/s)
3	Walking	-	1.2 (m/s)
4	Incline	4°	0.6 (m/s)
5	Incline	4°	1.2 (m/s)
6	Incline	9°	0.6 (m/s)
7	Incline	9°	1.2 (m/s)
8	Backwards	-	0.4 (m/s)
9	Backwards	-	0.7 (m/s)
10	Backwards	-	1 (m/s)
11	Walking	-	1.2 (m/s)
12	Running	-	1.8 (m/s)
13	Running	-	2.2 (m/s)
14	Running	-	2.7 (m/s)
15	Cycling	Resistance 1	70 (rpm)
16	Cycling	Resistance 3	70 (rpm)
17	Cycling	Resistance 5	70 (rpm)
18	Cycling	Resistance 1	100 (rpm)
19	Stairs	60 (W)	-
20	Stairs	60 (W)	-
21	Stairs	60 (W)	-
22	Standing	-	-
23	Sitting	-	-

Table 3: Activity Codes Key

Subject	Sex	Height (m)	Weight (kg)	Age (years)
1	Male	1.85	63.49	30
2	Male	1.77	63.49	25
3	Female	1.73	71.2	24
4	Male	1.68	68.03	25
5	Male	1.83	68.03	24
6	Male	1.85	68.03	25
7	Male	1.85	95.24	24
8	Male	1.85	65.76	33
9	Male	1.73	68.93	37
10	Female	1.63	58.05	27

Table 4: Subject Demographics

Note: All subject demographics were self-reported